



Space mining and scientific knowledge of asteroids: Implications for policy in a context of national interest

Christopher Williams

ESSCA School of Management, 55 Quai Alphonse le Gallo, Boulogne-Billancourt, 92513, France

ARTICLE INFO

Keywords:

Space mining
Asteroids
Astronomy
National interest

ABSTRACT

Space mining is increasingly an area of intense competition between firms and nations. It is also an industry that relies on scientific knowledge of asteroids. Knowledge generated through scientific research of asteroids can contribute to the future competitive advantage of firms and hegemony of nations seeking to secure leadership in the space mining sector. This makes scientific knowledge of asteroids valuable to market participants, conferring an advantage for them when accessed in a privileged form. We discuss the implications for policy and research of scientific efforts to learn more about asteroids given heightened geo-political tensions and polarizing national interests on Earth. At the core of this is a moral tension between greater global openness in the interests of Mertonian science and greater secrecy in the interests of nations that fund scientific capabilities.

1. Introduction

Space mining has become increasingly prominent in the discourse on the exploitation of space [1–4]. Given the vast scale of the Universe, the focus has been on celestial bodies within our own solar system, including the moon and asteroids.¹ A specific focus has been placed on near Earth objects (NEOs), the identification of these being one of the scientific goals of the recent Vera C. Rubin telescope in Chile which came into use in 2025. Over 40,000 Near Earth Asteroids (NEAs) have been identified² (as of writing in 2026) and many are thought to be easier to access than the Moon [5]. NEOs have orbits closer to Earth's orbit and are within 1.3 Astronomical Units (AUs) from the Sun.³ Precious metals, minerals, liquids and structural elements are contained in these relatively close bodies, alongside water, helium-3 and other molecules that can sustain human life in space or are in high demand on Earth. Indeed, resource depletion and scarcity on Earth [6] has made these potential sources of supply a priority for the emerging space mining industry [7].

Direct revenues from extractive space mining were not expected in the short or medium terms [8]. Nevertheless, the space mining market reached \$2.05 billion USD in 2025 and is expected to reach \$5.42 billion USD by 2030⁴ and \$23 billion USD by 2038⁵, as the ability of firms to directly extract from these celestial objects improves. Entrepreneurial space mining companies had a checkered start: Deep Space Industries (US) became defunct in 2019, Planetary Resources (US) became defunct in 2019, Moon Express (US) shifted direction to commercial payload services in 2018, while others including ispace (Japan), Asteroid Mining Corporation (UK), Transastra (US) and Astroforge (US) are still in operation in 2026 as small-medium sized enterprises. To date, no company has successfully mined resources from asteroids and returned them to Earth on a commercial basis.⁶

The space mining industry faces many formidable and diverse challenges. These include issues relating to resource rights and ownership [9], foundational components of the legal and governance system associated with mining in space [2]. Scholars have raised concerns

E-mail address: christopher.williams@essca.fr.

¹ Space mining has a principal interest in asteroids over comets: An asteroid is a minor planet, a small rocky body orbiting the sun, many found between the orbits of Mars and Jupiter, typically orbiting every 3–6 years. There are between 700,000 and 1,700,000 asteroids equal or larger than 1 km across. A comet is a small body made of dust and ice that orbits the Sun from beyond the orbit of Neptune (short period comets that orbit in less than 200 years) or from the more distant Oort Cloud (long period comets that orbit up to once every 250,000 years); it is estimated that there could be as many as 1 trillion comets.

² <https://neo.ssa.esa.int/discovery-statistics> accessed 7 May 2026.

³ About 2/3 of all NEOs are thought to be asteroids, while the remaining 1/3 the remnants of comets.

⁴ <https://www.thebusinessresearchcompany.com/report/asteroid-mining-global-market-report> accessed 7 May 2026.

⁵ <https://www.mordorintelligence.com/industry-reports/space-mining-market-industry> accessed 19 June 2025.

⁶ The first ever attempt to retrieve samples from an asteroid was Japan's Hayabusa probe which launched in 2003. While eventually returning c. 1500 microscopic grains to Earth in 2010, it suffered problems that threatened the mission: reaction wheel failure, multiple propulsion-related problems, and loss of attitude control.

<https://doi.org/10.1016/j.spacepol.2026.101785>

Received 17 February 2026; Received in revised form 13 May 2026; Accepted 6 June 2026

0265-9646/© 2026 The Author. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

regarding security, the potential for conflict and the militarization of space in connection with mining [10]. And space mining is an area of industrial focus that has a national interest status for many developed countries [11]. In addition to these geo-political factors, there are also immense technical challenges associated with engineering for mining in micro-gravity and achieving the technological feat that is required [10]. This constellation of issues puts a focus on the scientific knowledge associated with space mining, and the scientific teams at the frontier of research on asteroids.

Space mining is a knowledge intensive industry, highly dependent on scientific knowledge of asteroids, as well as many other disciplines. Most astrophysics research has a purely scientific goal of advancing human understanding of how the universe came about, particularly in galactic and extra-galactic science.⁷ Planetary science is different in terms of scale, methods, and typical questions asked. It has a focus on objects within our solar system and bridges astronomy, geology, chemistry, and atmospheric science, and even astrobiology [13]. At the same time, international treaties – such as the Outer Space Treaty (1967) – have come about to deny property rights on objects within our solar system. Because of the geo-politically sensitive and technologically high-risk nature of the space mining industry, knowledge produced by planetary scientists can be extremely valuable to market participants. The knowledge production system in astrophysics has received increasing attention by scholars [14]. However, there has been little research on the characteristics of the scientific teams examining asteroids and whether these teams have been – or are likely to become – aligned with national interest. This is important to understand within the context of the evolving New Space paradigm [15] in which there is a struggle for power involving private sector companies, institutions, and nation states, and in which space is seen as a “disputed setting of vast business and strategic potential” ([16], p. 10). We argue a key issue – one that has been overlooked – is the role of scientific teams examining asteroids and the governance of knowledge they generate within the space mining context.

Scientific teams researching asteroids can be viewed as part of this power struggle. The scientific knowledge that they generate can be a source of power within New Space [16] because of the imperative for firms to identify asteroids that contain ore; the percentage is expected to be very low [17]. This raises the question about how scientific knowledge of asteroids is created, protected and owned. Science of Team Science (SciTS) can be used as a theoretical anchor to examine the role of science in this type of context [18–21]. SciTS examines formation and processes in scientific teams including how scientific teams have become larger and more geographically diverse over the decades [19,22]. It acknowledges how team science operates in a more inter-disciplinary way than in the past to address ever more complex problems, and how scientific teams are influenced by institutional factors from the context in which they are situated [18].

We discuss future policy implications as they relate to space mining and scientific knowledge of asteroids. Firstly, we reflect on the promise of asteroids for space mining. Secondly, we consider the role of science of asteroids, and how scientific knowledge generated using ground-based and space-based telescopes can help the strategic decisions and technological choices and investments made by actors. Thirdly, we discuss science teams within a context in which space mining is becoming a nationally prioritized industry for leading nations. Fourthly, we develop policy and research implications by drawing on Mertonian norms of science. These norms of science established functional principles for how science ideally should work as a system, and how it can be protected from social or political bias. They allow us to develop and contrast two archetypal ‘Futures’ for policy – one more idealistic, one arguably more realistic – as it relates to the role of planetary science in

space mining.

2. The promise of asteroids

Asteroids are expected to play a pivotal role in the evolution of the space economy because of their potential for the extraction of valuable minerals and metals, including gold, nickel, iron, platinum group metals and rare Earths. This can satisfy resource constraints and shortages on Earth [1,4], or can be used during in-space development [23]. They also hold potential as sources of water and for the melting of trapped ice [24], important if mankind is to extend its habitat for sustained periods of time outside of Earth orbit, i.e., moving from a planetary to a multi-planetary existence [25].

In terms of satisfying resource constraints on Earth, commentators note how space mining could have benefits in supporting manufacturing, construction and technology development on Earth. Zacny et al. note how platinum group metals are likely to be more valuable than rare Earths in the shorter-term [7], despite discourse around rare Earth supply limitations [26]. Cannon et al. note that some metals will not be economically viable to return to Earth (including iron and nickel), while others, such as the platinum group metals will be in high enough concentrations and grades on certain types of asteroids to be economically viable [23]. The authors note how some metals – structural/photovoltaic metals – are more likely to be useful for in-space activity. In-situ resource extraction and use are recognized as being needed if mankind is to become permanently established in space [27]. As an example, space sourced propellants are expected to yield lower cost advantages for in-space travel, compared to Earth sourced propellants [28].

The promise that asteroids hold in terms of water, i.e., through the melting of trapped ice, is also discussed.⁸ While the Earth's water deposit is only 0.2% of its mass, asteroids may contain up to 15% water by mass, locked in clay deposits [29]. As mankind seeks to expand its civilization into space, including colonies on the Moon, Mars, and in remote space stations, access to water will be needed for drinking, and for agriculture and soil irrigation of space-based farms. Water is a pre-requisite for life, and it is commonly held that small bodies within the solar system were the source for water on Earth through the late bombardment epoch [30]. Accessing water from bodies in near Earth orbits will be necessary to support the expansion of human civilization into space.

Scholars have argued that space mining runs the risk of fueling inequality on Earth [4,26]. Some have called for new governance and legal infrastructure to deal with conflicts that could arise through space extraction [1,24], including international cooperation and formation of asteroid mining cartels [9] and the integration of AI into space resource management [2]. Others have argued how inequality could be managed by establishing practices that are sustainable in environmental, economic and social terms [31]. Butkeviciene and Rabitz note how space mining could reduce harmful economic and social effects on Earth by moving mining into space and therefore contributing to sustainable development on Earth [1].

3. The role of scientific knowledge for space mining

The economic promise of asteroids is linked to knowledge generated through space research, i.e., knowledge of opportunities (which asteroids), as well as knowledge that can be used in technological solutions for exploiting those opportunities. This knowledge is created using ground-based telescopes such as the Rubin Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory in Chile, and space-based telescopes such as NEOWISE (NASA), the Hubble Space Telescope (HST) and the James Webb Space Telescope (JWST). This scientific knowledge

⁷ In Banerke's (2020: 187) words, astronomy has “no practical application” [12].

⁸ Comets are thought to hold a higher abundance of interstellar material [28].

is highly valuable because of the extraordinary technological challenges involved. Terrestrial mining is itself highly risky, requiring in-depth scientific analysis and ingenuity to reduce geological uncertainty [32]. This fundamental mining risk is amplified with space mining; indeed, technological risks increase as the operating environment becomes an order of magnitude more hostile and extreme. The mining model used on Earth emphasizes heavy machinery and in-situ equipment and resources [33]. For space mining, equipment needs to be transported into space to land securely on fast moving and hostile objects, conduct mining operations autonomously, and then dispatch mined substances for transportation back to Earth or onward locations as the objects move again closer to Earth or to those locations. The state of technology available to do this has been described as ‘immature’ [4] and ‘not technically feasible’ [26], while the nature of the objects on which the mining would take place has been described as ‘poorly known’ [30].

These formidable issues raise the importance of science and research at the frontier of knowledge about asteroids if space mining is to move beyond scouting missions and fulfil its promise. All mining is based on sound modelling and understanding of the geological features of the ground [32]. And all space activity is scientifically rooted in the physical properties and laws of the universe. Without the use of telescope assets in the field of planetary science and its collaboration with many other disciplines including geology, chemistry and mathematics, the economic promise of space mining cannot be realized. This is even more pertinent given that asteroids have vastly different mineral and chemical compositions [17] as well as distinct surface morphologies; small bodies in our solar system are highly diverse in terms of shape, density, albedo, gravity and porosity [30]. Any physical landing comes with enormous risk of failure. This happened to the Philae lander in 2014 as it attempted to land and hook on to the comet 67P/Churyumov–Gerasimenko with harpoons, only to bounce and end up on its side in a dark crevice. Assumptions about the characteristics of the object’s surface had proven to be incorrect, although the project did create scientific impact [34].

While scientific knowledge of asteroids in our solar system was criticized as being ‘poor’ less than 15 years ago [30], the stock of this knowledge has expanded over time, particularly as newer and increasingly sophisticated observational platforms have been made available to astronomers operating in different wavelengths (optical, infrared, x-ray, radio). Fig. 1 shows the distribution of publications from the NASA Astrophysics Data System operated by the Smithsonian Astrophysical Observatory on “Asteroids” over the period 1990–2026 ($n = 49,475$).⁹ This shows a steady increase in scientific productivity over time, with a yearly upward trend for refereed publications.

Competing firms in the space mining sector will use scientific knowledge to inform and guide their strategic decisions and investments. This knowledge guides the choices made by space mining companies at a technological level, as well as in terms of strategy, including market segmentation decisions, financing and the management of risk.

Firstly, it will help them in their patenting decisions, a crucial form of intellectual property protection in this industry [16,35] as equipment is adapted and invented to deal with the conditions on asteroids. Secondly, scientific knowledge will help space mining companies make decisions about targeting: which asteroids to target for missions. While over 40,000 NEAs have been identified, only a small fraction is economically viable. According to one analysis, there are less than 20 water-ore-bearing NEOs >100m diameter [17]. Competitors need to be highly selective and focused on minimizing risk and maximizing returns. Scientific techniques such as spectroscopy allow asteroids to be classified in terms of chemical composition [36], with observations from space-based telescopes at longer infrared wavelengths than is possible with ground-based telescopes able to discern large scale heterogeneities. While these cannot resolve small spatial features or see beneath the

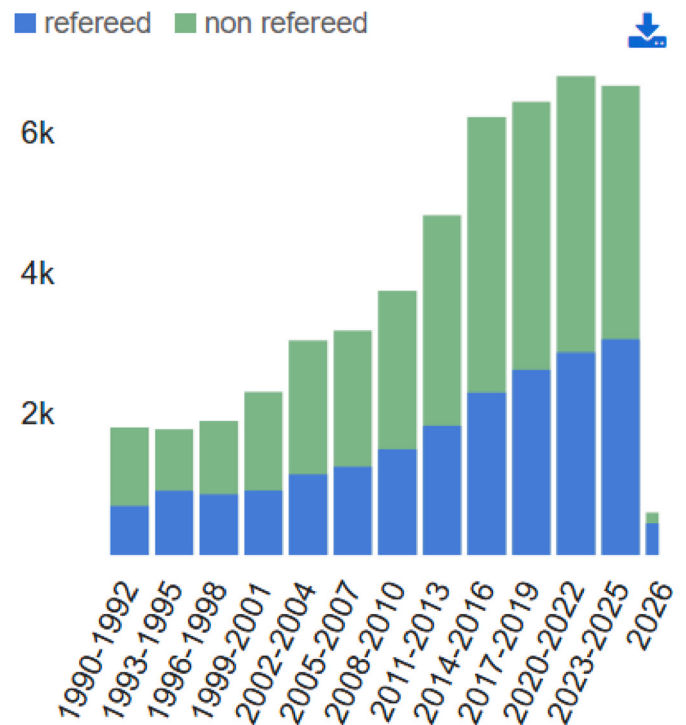


Fig. 1. Results from Harvard's Astrophysics Data System on published articles on “Asteroids” (1990–2025) ($n = 49,475$) (May 2026).

Source: Author query on <https://ui.adsabs.harvard.edu/>

surface, high-resolution precision imaging and thermophysical analysis can be combined to aid understanding of porosity, grain size and surface stability. These critical elements of knowledge can support companies in prioritizing targets. The experience of OSIRIS-REx on Bennu in 2019 showed that unexpected detail can arise only once the spacecraft arrives at the asteroid [37]. This included the discovery that the asteroid was active, ejecting particles from its surface, and requiring new scientific analysis of locally captured data [37]. Coarse scientific knowledge is a key part of strategic planning; choosing celestial bodies, and developing business cases that involve many partners, including payload companies, and the investment community. More refined and accurate scientific knowledge becomes available as missions approach targets and is essential for managing safety and operational risks.

4. Science teams within an emerging national priority context

The scientific research relevant to progress in space mining emanates from many teams made up of scientists spread across the countries of the world. Scientific knowledge of asteroids is created by these teams operating within an institutional context that is changing rapidly and becoming more complex. Modell discusses three sets of institutional logics that have emerged within space mining: entrepreneurial (a market capitalism logic), sustainability (a cooperative capitalism logic), and regulatory (a state capitalism logic) [3]. Within the regulatory logic, there have been both transnational and national-level efforts, reflecting changing geo-political forces [3]. Transnational efforts have attempted to establish norms relating to the creation and sharing of scientific knowledge of space. Building on the Outer Space Treaty, Section 8 of the Artemis Accords establishes a clear norm that scientific data generated through civil space exploration activities should be shared openly, promptly, and for the benefit of all humankind. However, it does not specify immediate, real-time release of raw scientific data. And it does not force disclosure of commercial or proprietary data. The United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) Space Resources Working Group has argued for transparency and

⁹ Query made in May 2026.

cooperation, although its mandate is limited when it comes to the role of scientific knowledge. Open scientific data is far more explicitly articulated in Artemis than in UNCOPUOS processes.

Meanwhile, several powerful countries have designated space mining an industry of national strategic importance and have enacted laws that explicitly recognize property rights over resources extracted. As of 2026, these include the US with its Commercial Space Launch Competitiveness Act (2015), Luxembourg with its Space Resources Act (2017) and Japan with its Act on the Promotion of Business Activities for Exploring and Developing Space Resources (2021). Other countries have enacted national laws with relevance to space mining without making provisions for ownership of extracted resources, such as the United Arab Emirates with its Federal Law no. 12 (2019), Brazil with its Space Activities Act (2024), and Italy with its Space Economy Law (2025). Other powerful space sector countries have signaled national interest and strategic intent with respect to the space mining sector. President Xi Jinping was quoted as saying: “Chang’e-6 collected samples from the moon’s far side for the first time in human history and returned to Earth, marking another landmark achievement in China’s endeavor to build a strong country in space as well as in science and technology ... Each bold idea and its successful implementation in lunar exploration is a full demonstration of the human capacity to gain knowledge of and utilize a celestial body”¹⁰

It is within this environment of mixed transnational efforts and emerging national interests that science teams form to study asteroids; the resulting knowledge being potentially valuable to nationally prioritized industries, especially scientific knowledge that contains early insights that are protected from public dissemination for an exclusivity period. Table 1 shows a list of the accepted scientific General Observer (GO) programs on the James Webb Space Telescope (JWST) over Cycles 1 – 5 (2021–2026) that were targeted on the study of asteroids. It is interesting to look at JWST in this context for three reasons. Firstly, it is the most powerful and expensive observational platform ever built (at the time of writing) and was primarily US-funded (Canadian and European agencies contributed less than 20% combined). Consequently, it was very difficult for astronomers to gain access to; Cycle 5 had an acceptance rate of ~9%. Any programs accepted to use the platform to study asteroids would be expected to exhibit a high level of scientific merit, competing against proposals to study phenomena much deeper in space. Secondly, it was a recent platform at the time of writing, and its usage took place squarely within the era of the New Space paradigm. Thirdly, information about its use, including details of investigators and the scientific nature of their programs is openly available. This would not be possible with platforms based in some other countries.

Table 1 shows GO programs, i.e., the competitive, open programs for observation time, that were in the Solar System Astronomy scientific category and that were specifically targeted at asteroids, according to the program description information. It excludes programs that targeted asteroid-linked phenomena in the category of exoplanets – such as exoasteroids - (i.e., beyond our solar system) and excludes studies of trans-Neptunian objects (TNOs) (icy bodies beyond Neptune). In keeping with the purpose and capabilities of JWST to reveal objects deep in space and time, the number of programs and total observing time on asteroids is very small compared to the number of overall accepted programs. The mean observing time for the 14 programs listed that were not archival research was 28.1 h, slightly less than the JWST mean. However, this was skewed by program 11793 which was allocated a massive 153 h, mandating a zero exclusive access period. We note the following.

- all accepted asteroid programs were led by a Principal Investigator (PI) based at a US institution;

- all the investigating teams were US dominated;
- the majority (12/19) had investigating teams made up of scientists entirely from US institutions;
- the vast majority requested the full exclusive access period over results (12 months) for the programs where this was applicable (13/14);
- roughly half of the programs (9/19) had an investigator from a US-based private or commercial entity.

This profile could be suggestive of a national interest bias in this category of science programs, although it is not proof of a pervasive national interest bias within the international asteroid science community.

The potential for a relationship between the characteristics of science teams working on asteroid research and national interest can be explored using the Science of Team Science (SciTS) perspective [18,19,21]. SciTS is an interdisciplinary research field that studies how scientific teams form, function, perform, and produce knowledge, especially in large, distributed, interdisciplinary, and mission-oriented contexts. A central tenet of SciTS is that novelty and impact in modern science are created by purpose-driven teams of collaborating scientists; the prominence of solo authors has diminished over time [22,38]. It stresses the importance of characteristics of science teams for understanding their purpose, governance, and performance. Such characteristics include the size of the team [39], the diversity (including degree of internationalization) of the team [18], and how the team is lead [20]. In the case of asteroid research using JWST shown in Table 1, the teams are generally small and compact, with low or zero international diversity, and they are all US-led. Geographic bias is emphasized in SciTS literature, as it can colour the perception of quality of scientific studies [40]. SciTS literature argues that larger team sizes are associated with greater openness [41], as well as higher impact [22,35]. Smaller teams that are low in geographic diversity will incur lower coordination costs, a potential consideration in a commercial and national interest context. The SciTS perspective examines how these types of factors contribute to productivity and impact within a scientific field.

Importantly for space mining and the science of asteroids, SciTS also emphasizes the institutional environment in which science teams form and operate [21]. This environment determines structure, policy and culture that influence how science teams perform [18]. In other words, the norms, legitimacy and incentives used in the formation of science teams are derived from the institutional context, including norms and expectations related to team size, degree of multi-nationality of the team, and who leads the team. The institutional environment does not only provide norms, but it also provides tangible resources. This includes funding and the built environment [18]. In this sense, institutions place boundary conditions on the formation, purpose and performance of scientific teams. In the case of science teams working on asteroid research, this institutional context includes the country (or countries) funding and building telescope assets, the country and institution(s) responsible for operating telescope assets, and the governance of processes for deciding who gets to use those assets for generating new knowledge of asteroids.

In the United States, the National Science Foundation (NSF) is the lead institution for ground-based astronomy and has primary responsibility for funding and operating ground-based astronomy facilities, while NASA has primary responsibility for space-based observatories. However, there are important exceptions and partnership arrangements such as NASA’s funding for research programs using ground-based data, and non-US institutional involvement in the development of space-based telescopes such as JWST. For several flagship NASA space telescopes (notably Hubble, JWST, and Roman), NASA contracts the non-profit Association of Universities for Research in Astronomy, Inc. (AURA) to operate the missions through STScI, which manages science operations and peer-reviewed time allocation. AURA is a university consortium that operates major astronomical observatories

¹⁰ <https://news.cgtn.com/news/2024-09-23/Xi-Jinping-s-key-quotes-on-China-s-lunar-exploration-1x8rnUMMioE/p.html>.

Table 1

JWST Cycles 1 – 5 General Observer (GO) programs studying asteroids

Source: Author queries on <https://www.stsci.edu/jwst/science-execution/program-information>.

Cycle	Program Id (STScI)	Title	Exclusive Access Period (max. allowed = 12 months)	PI Institution	Co-I Institutions	Commercial or Private Entity listed as investigating institution?
1	1731	Variability and Abundance of Hydration on M-Type Asteroid (16) Psyche: The Search for Water on the Largest Metallic Asteroid	12	USA	All USA	Yes: two from Southwest Research Institute: a private non-profit organization
1	2211	A Pure Parallel Survey of Water in the Asteroid Belt	12	USA	6/7 USA	No
1	2361	Are L-type Asteroids Ancient Reservoirs for the Earliest Solar System Solids?	12	USA	All USA	No
1	2537	Analysis of Low-Albedo Asteroids Observed through JWST Guaranteed Time	12	USA	4/5 USA	No
2	3195	Are M-Type asteroids the remnant core fragments of the planetesimals?	12	USA	All USA	No
2	3760	Investigating the cause of the 3 μ m absorption feature on asteroids	12	USA	All USA	Yes: two from Southwest Research Institute: a private non-profit organization
3	6384	Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar System	12	USA	5/9 USA	No
4	7937	Cometary and Asteroidal Dust in our Zodiacal Cloud from Archival MIRI Observations of the 10- μ m Silicate Feature	AR	USA	All USA	No
4	8214	Mining JWST data for hidden asteroid gems	AR	USA	5/10 USA	Yes: Parrott Studio, LLC, a private, for-profit company, listed as an investigator institution
4	8663	Is the Moon an Active Source of Near-Earth Asteroids? Testing the Origins of Earth Quasi-Satellite Kamo'oalewa	12	USA	All USA	No
4	8782	Nominally anhydrous asteroids as reservoirs of water in the inner solar system	12	USA	All USA	No
4	8812	Distribution of water in the solar system from crossover region spectroscopy of asteroids	AR	USA	All USA	Yes: one (solo investigator) from Southwest Research Institute: a private non-profit organization
5	9903	Preparing for the Worst: A ToO Program for Utilizing JWST for Critical Characterization of Potential Impactors	0 (Program may not be triggered)	USA	11/16 USA	No
5	9938	A martian origin for olivine-dominated small bodies	12	USA	All USA	Yes: one investigator from Fibernetics LLC, a private, for-profit U.S. manufacturer of fiber-optic-gyroscope (FOG)-based inertial measurement units (IMUs)
5	10012	Mapping water in the Solar System with asteroids in the JWST public archive	AR	USA	All USA	No
5	11464	A spectroscopic test of the origins of Earth co-orbitals with JWST/NIRSpec	12	USA	6/9 USA	Yes: PI from Eureka Scientific Inc., a private non-profit organization
5	11776	Mysterious Origins of D-Type Asteroids	12	USA	All USA	Yes: one investigator from Southwest Research Institute and one investigator from Fibernetics LLC
5	11793	The Composition and Origins of the Main-Belt Asteroids	0	USA	8/9 USA	Yes: one investigator from Fibernetics LLC, a private, for-profit
5	12108	Probing the origins and evolution of water-rich asteroids using the targets of the Emirates Mission to the Asteroid belt	12	USA	All USA	Yes: one from Southwest Research Institute: a private non-profit organization

and science institutes for NSF and NASA under federal contract. This is all part of the institutional environment in which science teams form for asteroid research.

This institutional context differs across the regions of the world. For instance, funding is more fragmented in Europe compared to the US, with many national agencies (CNRS, DFG, UKRI, INFN) as well as European-level bodies (ESO, ESA, ERC). While there is very strong institutional support in the US, this is more moderate in Europe [42]. In terms of asset governance, the centralized time allocation committees (TACs) found in the US are less common in Europe where there is a greater pressure for guaranteed national shares, and arguably more complex access rules [43]. In China, the Chinese Academy of Sciences (CAS) is the scientific authority and owner of space telescope missions, including proposal evaluation and time allocation. A large share of telescope time is pre-allocated, and there is less formal separation between mission builders and users [44]. A high priority is given to researchers at CAS institutes.

These different institutional contexts raise a policy question in terms

of how to regulate the formation and operation of science teams in asteroid research, and what the guiding principles are for whether knowledge created is aligned with national strategic priorities and private sector progress in space mining.

5. Policy and research implications

This policy question can be framed using the concept of normative structure in science that was introduced by Robert K. Merton [45] and famously contested by Ian Mitroff [46]. Merton described four institutional imperatives – commun(al)ism (that science advocates anti-secrecy and intellectual property should be shared), universalism (that there are no biases in the search for truth based on personal characteristics of scientists or international boundaries), disinterestedness (scientists serve the interests of the scientific community directly, rather than their personal benefit), and organized skepticism (the “temporary suspension of judgement”; claims are subjected to critical scrutiny). These norms have been contested over time [46]. However,

Mertonian norms are still useful as analytical reference points; they should not be seen as what scientists actually or always do and may be seen as empirically unrealistic. But they remain valuable as ideals, as what science might hold important in terms of institutional values [47]. In the modern era, we see the Open Data Movement as supporting the norm of commun(al)ism. The use of blinded and dual-anonymized reviews supports universalism. Conflict of interest rules are aligned with disinterestedness and the culture of peer review in science aligns with organized skepticism. Policy and future research implications can be developed by framing science of asteroids according to Mertonian norms and considering the implications of any trend away from these norms in practice.

Table 2 depicts two alternative futures for the generation of scientific knowledge on asteroids. In Future 1, Mertonian norms are upheld. Supporting the norm of commun(al)ism, all scientific knowledge generated on asteroids is shared publicly and internationally, for the benefit of all, including market participants from any jurisdiction, without any exclusive access period. All scientists, regardless of country of origin, or country of telescope, would subscribe to this, and this would be consistent with Section 8 of the Artemis Accords. Supporting the norm of universalism, scientific teams are not formed with any bias, including nationality or institution bias, and teams are likely to be larger and more internationally diverse than the alternative, Future 2 below, consistent with this norm. Supporting the norm of disinterestedness, scientific efforts serve science and not any interests of private sector or commercial actors. And supporting organized skepticism, the global community of scientists is responsible for continually scrutinizing the research analysis conducted.

At the opposite end of the spectrum we have in Future 2 a situation where commercial interests and national priorities have a far greater influence. There is a higher emphasis on secrecy to privilege market participants and allow them to gain an advantage through technological

and strategic decisions guided by scientific research. This is closer to what Mitroff termed solitariness [46], a counter-norm to commun(al)ism, as scientists attempt to secure priority to underpin temporary competitive advantage. Scientific teams are more likely to be comprised of individuals emanating from one country: the country owning and operating the telescope in question. The quality of work is evaluated based in part on the identity of these individuals, and judgement is not impersonal. Mitroff referred to this as particularism, a counter-norm to universalism. In Future 2, scientists are more motivated by personal interests, including private sector incentives and/or a sense of national pride or identity. These personal interests shape their scientific behavior, and this interestedness – in Mitroff's terms – is a counter-norm to the pure quest for knowledge as reflected in Merton's disinterestedness. In Future 2, scrutiny of scientific knowledge is increasingly done by the private sector and government agencies with an interest in space mining and acting in a national interest. It may entail what Mitroff termed organized dogmatism – a counter-norm to Merton's organized skepticism – when there is a prolonged commitment to sustained assumptions and research programs.

Futures 1 and 2 represent archetypal or stylistic scenarios that may never occur in their purest or most simplistic form. While Future 1 can be seen as normative, Future 2 may be closer to what is emerging in practice. This echoes the Merton – Mitroff contrast as documented many decades ago. The two Futures can be considered as two ends of a spectrum. Policy and research relating to the role of science in space mining can be guided by the continuum of positions on this spectrum.

5.1. Policy implications

Firstly, policy needs to understand and anticipate the nature of shifts between Futures 1 and 2, i.e., between the ideal for freedom and openness of science (Future 1 in Table 2) and pressures for privileged knowledge in the national interest (Future 2 in Table 2). Commentators have stressed the power struggles and structural hegemony of a few dominant states in the space economy. According to Rementeria: "... powerful governments will go to great lengths to preserve their heritage and secure the advantage obtained by public space agencies through decades of sustained investment and practical know-how acquisition" [16, p. 10]. This suggests frontier knowledge generation on asteroids is likely to be pulled towards Future 2 in the medium to longer term. In this scenario, scientific knowledge generated through the use of a research asset will be concentrated in the country that funds the asset's development and operation, consistent with this structural hegemony view. This may result in reinforcing the scientific divide in astronomy research between richer and poorer countries by creating a barrier that prevents researchers from poorer countries accessing networks, platforms, and resources [48] and could translate over time into economic inequality in later stages of the development of the space economy, a fear noted in space policy literature [4,26].

In Future 2, there will be higher centralization and tighter control of asteroid science programs. There has been a long-term trend in astronomy towards higher team sizes [49,50], consistent with the SciTS perspective and science teams at large [19,22]. Future 2 suggests asteroid research will buck this trend and stand in contrast with SciTS in terms of the larger team assumption [41]. Future 2 will be characterized by smaller, more homogeneous teams, which, interestingly, have been seen to contribute to disruptive breakthroughs in astronomy [42]. Smaller science team sizes would be indicative of lower coordination and communication costs [51,52], and tighter control. This is consistent with a restriction of the outward diffusion of newly generated knowledge, making the knowledge more economically valuable for actors with access to it. Future 2 will see greater secrecy in asteroid science. While policy has encouraged astronomers to embrace greater openness and to release data and findings early [53], astronomers can still request a maximum exclusive access period within the rules of a given telescope operator. If scientists involved are aware of the value of their results for

Table 2
Contrasting futures of openness versus secrecy in asteroid research.

Mertonian norm	Future 1 – Mertonian-based open science for asteroid research	Future 2 – A closed, privileged system for asteroid research
Commun(al)ism (science advocates anti-secrecy and intellectual property should be shared)	Widespread open sharing of all scientific results on asteroids across international boundaries.	Scientific projects are aligned with commercial interests of one or a few select private sector companies and individual nations. Results are kept (temporarily) secret for the benefit of the private sector and/or a nation state.
Universalism (there are no biases in the search for truth based on personal characteristics of scientists or international boundaries)	Scientific teams for asteroid research drawn from diverse countries and backgrounds.	Scientific teams for asteroid research are biased and drawn from the country owning and operating the telescope in question.
Disinterestedness (scientists serve the interests of the scientific community directly, rather than their personal benefit)	Scientific research on asteroids is conducted for scientific audiences only; there is no commercial gain for individuals involved or links to the private sector or national incentives.	Scientists involved gain tangible benefits (e.g., financial rewards) or intangible benefits (e.g., a sense of national pride) through the work they conduct on asteroids.
Organized skepticism (the "temporary suspension of judgement"; claims are subjected to critical scrutiny)	Scientific peer review continues to be conducted for all results, using peers across the global community of scientists.	Private sector commercial actors play a prominent role in evaluating the results of research studies on asteroids.

the private sector and national interests, they may come under pressure to protect priority for their efforts for as long as possible.

Science teams are often taken for granted in space policy literature; more attention has been on issues relating to governance, law and rights, sustainable development, and business models [1,8,15,24]. SciTS logic can be useful here, especially its emphasis on how the institutional environment determines science team composition and performance. Oftentimes, the relevant institutional forces are nationally or regionally defined, as the contrast between the US and European environments described above indicates [42]. Space policy vis-à-vis space mining should consider how scientific knowledge is generated across the Future 1 – Future 2 spectrum, whether the teams involved are globally open and unbiased and in line with Mertonian norms of science, or whether they operate in a tightly controlled way in one single country, emphasizing secrecy and priority in a national interest. Stifling the diffusion of scientific knowledge of asteroids may reinforce a shift towards Future 2 and the hegemony of a few powerful countries. Policy needs to incorporate how the utilization of publicly funded research assets is governed, monitored and assessed. The SciTS view suggests this needs to involve attention to the social structures of scientific teams that generate this knowledge. This relates not just to characteristics such as their size and degree of geographic diversity, but also how authority is enacted through hierarchical relationships in smaller teams versus large, distributed networks.

Relatedly, space policy needs to consider the role played by asset controllers, i.e., the organizations that control scientific asset utilization. Depending on the country, this is not necessarily the same as asset builders. These organizations exert power as gatekeepers in deciding access rights to platforms for scientists to generate knowledge on asteroids. They also establish asset level policy concerning the extent to which this knowledge is kept exclusive, and for how long. Asset controllers are a critical part of the institutional system in scientific communities, and they work within national contexts. Future 2 may create an isomorphic effect across space mining nations that own and operate telescope assets and have established scientific communities in asteroid research. Such an effect would be localized islands of policy for knowledge governance and appropriation that is embodied in formal and informal (cultural) norms for science in each nation's own interest.

5.2. Research implications and avenues

Secondly, there are implications of the perspective taken here for future research. Research is needed to examine the interface between space mining companies and sources of scientific knowledge, including how companies access and internalize scientific knowledge and how this process is governed. It should examine how this knowledge is used within firms to generate new intellectual property and firm-specific advantages based, for instance, on improvements to space mining technology or launch and communication systems. It can also explore how scientific knowledge informs strategy and strategic business decisions for the companies involved. The latter includes decisions relating to market segmentation and targeting: which category of object to target and then which specific objects to give priority to. In terms of categories, C-type [carbonaceous] are water rich; L-type refer to low-albedo asteroids that are expected to hold little economic value; S-type [silicaceous] are silicate dominated and pose challenges in terms of separation of metals; and M-type [metallic] asteroids are mostly dry and high in metal content. Research can look at how scientific knowledge informs market segmentation for space mining companies, and how they use science in the processes of planning, financing and risk management, public communications and management of stakeholders. Scientific knowledge could have a role to play in guiding each of these strategic processes, and this could be examined through case studies.

Future research can also provide insight into the links between scientific work on asteroids and national priority by comparing the composition of scientific teams and their performance across a range of

telescopes, including ground-based and space-based telescopes and those run by American, European, Japanese, Russian, Indian and Chinese agencies. By comparing science team characteristics using telescopes built and operated in different national contexts, it will be possible to appreciate the role played by national interest in valuing scientific knowledge of asteroids. This line of research can explore the perceptions and views of the scientific community and the extent to which international networks in science are aligned with national interests.

One of the stated goals of the Vera C. Rubin telescope in Chile is to identify new NEOs, as well as other – potentially minable – objects in the solar system. It would be interesting to examine how space mining companies within the New Space paradigm develop and use new capabilities to examine the vast amount of data produced by survey telescopes on minable objects over longer periods of time. This will be a vast data analysis exercise that will lend itself to artificial intelligence. How companies use this technology to compete and gain advantage based on vast data sets of publicly available scientific knowledge will be useful to study as it may reveal the extent to which the field is moving towards or away from Future 2.

6. Conclusion

Scientific knowledge of asteroids is – and will continue to be – highly valuable to market participants in space mining, including knowledge of their physical and geological properties that can help in evaluating their status as minable objects. Indeed, a plethora of new scientific knowledge is currently emerging on asteroids thanks to new space telescopes including new ground-based telescopes such as the Vera C. Rubin telescope in Chile and the planned NEO Surveyor space-based telescope. Nevertheless, recent literature on the New Space paradigm does not give a prominent position to the role that frontier science plays in enabling the space mining economy, focusing understandably on institutional dynamics [3] and downstream market-pull factors. Upstream scientific work is a critical part of the policy context for space mining and is not disconnected from international power-struggles that are set to intensify. This is especially the case as countries gear up for actual resource exploitation, i.e., to go beyond scouting missions. Whether scientific teams in asteroid research adhere to the ideals of Mertonian norms versus supporting a paradigm of national interest and protectionism will become clearer over the coming years.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] E. Butkevičienė, F. Rabitz, Sharing the benefits of asteroid mining, *Global Policy* 13 (2) (2022) 247–258.
- [2] J. García-del-Real, M. Alcaráz, Unlocking the future of space resource management through satellite remote sensing and AI integration, *Resour. Policy* 91 (2024) 104947.
- [3] S. Modell, Space mining as an emerging organisational field: an institutional logics perspective, *Space Policy* 74 (2025).
- [4] J.A. Peña-Ramos, F.R.R.D. Luis, Resources in space and asteroid mining: where we are and which challenges should be expected, *Int. J. Technol. Manag.* 82 (3–4) (2020) 197–205.
- [5] E. Perozzi, R. Binzel, A. Rossi, G.B. Valsecchi, Asteroids more accessible than the moon, *EPSC* 5 (2010) 750.
- [6] V. Balaram, Potential future alternative resources for rare earth elements: opportunities and challenges, *Minerals* 13 (3) (2023) 425.

- [7] K. Zacny, M.M. Cohen, W.W. James, B. Hilscher, Asteroid mining, in: AIAA Space 2013 Conference and Exposition, 2013, p. 5304.
- [8] M. Weinzierl, Space, the final economic frontier, *J. Econ. Perspect.* 32 (2) (2018) 173–192.
- [9] N. Zhang, B. Liu, H. Lin, Long live equity! division of labor and distribution in asteroid mining. Division of Labor and Distribution in Asteroid Mining, 2022 (February 24, 2022).
- [10] O. Steffen, Explore to exploit: a data-centred approach to space mining regulation, *Space Policy* 59 (2022) 101459.
- [11] M. Borowitz, A. Noonan, R. El Ghazal, US strategic interest in the moon: an assessment of economic, national security, and geopolitical drivers, *Space Policy* 69 (2024) 101548.
- [12] D. Baneke, Let's not talk about science: the normalization of big science and the moral economy of modern astronomy, *Sci. Technol. Hum. Val.* 45 (1) (2020) 164–194.
- [13] D. Reinecke, J. Bimm, The maintenance of ambiguity in Martian exobiology, *Soc. Stud. Sci.* 52 (2) (2022) 199–226.
- [14] R. Heidler, Cognitive and social structure of the elite collaboration network of astrophysics: a case study on shifting network structures, *Minerva* 49 (4) (2011) 461–488.
- [15] K. Bousedra, Downstream space activities in the new space era: Paradigm shift and evaluation challenges, *Space Policy* 64 (2023) 101553.
- [16] S. Rementeria, Power dynamics in the age of space commercialisation, *Space Policy* 60 (2022) 101472.
- [17] M. Elvis, How many ore-bearing asteroids? *Planet. Space Sci.* 91 (2014) 20–26.
- [18] K.L. Hall, A.L. Vogel, G.C. Huang, K.J. Serrano, E.L. Rice, S.P. Tsakraklides, S. M. Fiore, The science of team science: a review of the empirical evidence and research gaps on collaboration in science, *Am. Psychol.* 73 (4) (2018) 532.
- [19] D. Stokols, K.L. Hall, B.K. Taylor, R.P. Moser, The science of team science: overview of the field and introduction to the supplement, *Am. J. Prev. Med.* 35 (2) (2008) S77–S89.
- [20] S. Yu, H.D. Bedru, I. Lee, F. Xia, Science of scientific team science: a survey, *Computer Science Review* 31 (2019) 72–83.
- [21] K. Börner, N. Contractor, H.J. Falk-Krzesinski, S.M. Fiore, K.L. Hall, J. Keyton, B. Uzzi, A multi-level systems perspective for the science of team science, *Sci. Transl. Med.* 2 (49) (2010), 49cm24.
- [22] B. Uzzi, S. Mukherjee, M. Stringer, B. Jones, Atypical combinations and scientific impact, *Science* 342 (6157) (2013) 468–472.
- [23] K.M. Cannon, M. Gialich, J. Acain, Precious and structural metals on asteroids, *Planet. Space Sci.* 225 (2023) 105608.
- [24] M. Ingebreetsen, Mining asteroids, *IEEE Spectr.* 38 (8) (2001) 34–39.
- [25] X.S. Yap, R.E. Kim, Towards earth-space governance in a multi-planetary era, *Earth System Governance* 16 (2023) 100173.
- [26] M. Feinman, Mining the final frontier: keeping Earth's asteroid mining ventures from becoming the next gold rush, *Pitt. J. Tech. L. & Pol'y* 14 (2013) 202.
- [27] P.T. Metzger, Space development and space science together, an historic opportunity, *Space Policy* 37 (2016) 77–91.
- [28] F. Xu, The approach to sustainable space mining: issues, challenges, and solutions, in: IOP Conference Series: Materials Science and Engineering, vol. 738, IOP Publishing, 2020 012014, 1.
- [29] C.M.O.D. Alexander, K.D. McKeegan, K. Altwegg, Water reservoirs in small planetary bodies: meteorites, asteroids, and comets, *Space Sci. Rev.* 214 (1) (2018) 36.
- [30] M.A. Barucci, E. Dotto, A.C. Levasseur-Regourd, Space missions to small bodies: asteroids and cometary nuclei, *Astron. Astrophys. Rev.* 19 (2011) 1–29.
- [31] J.A. Dallas, S. Raval, J.A. Gaitan, S. Saydam, A.G. Dempster, Mining beyond earth for sustainable development: will humanity benefit from resource extraction in outer space? *Acta Astronaut.* 167 (2020) 181–188.
- [32] S.X. Li, P. Knights, D. Dunn, Geological uncertainty and risk: implications for the viability of mining projects, *J. Coal Sci. Eng.* 14 (2008) 176–180.
- [33] S. Saydam, Extracting off-earth resources. Extracting Innovations: Mining, Energy, and Technological Change in the Digital Age, 2018, pp. 207–220.
- [34] L. O'Rourke, P. Heinisch, J. Blum, S. Fornasier, G. Filacchione, H. Van Hoang, H. Sierks, The Philae lander reveals low-strength primitive ice inside cometary boulders, *Nature* 586 (7831) (2020) 697–701.
- [35] M.E. Wachowicz, M. Bury, Space technology transfer problems in the context of protecting the space heritage, *Space Policy* 42 (2017) 66–69.
- [36] T.H. Burbine, T.J. McCoy, E. Jaresowich, J.M. Sunshine, Deriving asteroid mineralogies from reflectance spectra: implications for the MUSES-C target asteroid, *Antarct. Meteor. Res.* 16 (2003) 185–195, 16, 185–195.
- [37] K.S. Lauretta, D.N. DellaGiustina, C.A. Bennett, D.R. Golish, K.J. Becker, S. Balram-Knutson, C.W.V. Wolner, The unexpected surface of asteroid (101955) Bennu, *Nature* 568 (7750) (2019) 55–60.
- [38] S. Wuchty, B.F. Jones, B. Uzzi, The increasing dominance of teams in production of knowledge, *Science* 316 (5827) (2007) 1036–1039.
- [39] Y.N. Lee, J.P. Walsh, J. Wang, Creativity in scientific teams: unpacking novelty and impact, *Res. Pol.* 44 (3) (2015) 684–697.
- [40] M. Kowal, P. Sorokowski, E. Kulczycki, A. Żelaźniewicz, The impact of geographical bias when judging scientific studies, *Scientometrics* (2022) 1–9.
- [41] K.R. Lakhani, L.B. Jeppesen, P.A. Lohse, J.A. Panetta, The Value of Openness in Scientific Problem Solving, Division of Research, Harvard Business School, Boston, MA, 2007, pp. 7–50.
- [42] T. Heinze, M. von der Heyden, D. Pithan, Institutional environments and breakthroughs in science. Comparison of France, Germany, the United Kingdom, and the United States, *PLoS One* 15 (9) (2020) e0239805.
- [43] ASTRONET, Science Vision and Infrastructure Roadmap 2022–2035, European Astronomical Community, 2023. https://www.astronet-eu.org/wp-content/uploads/2023/05/Astronet_RoadMap2022-2035_Interactive.pdf. (Accessed 8 May 2026).
- [44] Gong, Y., Miao, H., Zhan, H., Li, Z. Y., Shangguan, J., ... & Zu, Y. Introduction to the Chinese space Station Survey telescope (CSST). *Sci. China Phys. Mech. Astron.*, 69(3).
- [45] R.K. Merton, Science and technology in a democratic order, *Journal of Legal and Political Sociology* 1 (1) (1942) 115–126.
- [46] I.I. Mitroff, Norms and counter-norms in a select group of the Apollo moon scientists: a case study of the ambivalence of scientists, *Am. Sociol. Rev.* (1974) 579–595.
- [47] T.F. Gieryn, Boundary-work and the demarcation of science, *Am. Sociol. Rev.* 48 (6) (1983) 781–795.
- [48] G.R. Stahlman, Is there a scientific digital divide? Information seeking in the international context of astronomy research, in: International Conference on Information, Springer Nature Switzerland, Cham, 2023, March, pp. 514–523.
- [49] S. Milojević, Principles of scientific research team formation and evolution, *Proc. Natl. Acad. Sci.* 111 (11) (2014) 3984–3989.
- [50] J.R. Najita, Investing for discovery in astronomy, arXiv preprint arXiv:1907.11700 (2019).
- [51] A. Specht, K. Crowston, Interdisciplinary collaboration from diverse science teams can produce significant outcomes, *PLoS One* 17 (11) (2022) e0278043.
- [52] C.S. Wagner, T.A. Whetsell, S. Mukherjee, International research collaboration: novelty, conventionality, and atypicality in knowledge recombination, *Res. Pol.* 48 (5) (2019) 1260–1270.
- [53] C. Williams, Allocating time on scientific platforms in outer space: evidence from James Webb Space Telescope Cycle 1-3 general observer programs, *Res. Pol.* 54 (5) (2025) 105239.